



Network or Control?

A Validated Spectral Bridge for Inverter-Dominated Grids

Weak-node and weak-link analysis as an application of the controller-aware bridge

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Scan for code and data
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Not every weak node is weak for the same reason. Some need copper; some need control.

Practical consequence for IAS: A planner using the standard reduced model can approve an inverter conversion while the true critical control mode stays invisible.

1. The problem IAS recognizes

Industrial grids and microgrids are replacing synchronous machines with inverter-based resources. The visible change is lower inertia. The subtle change is damping location.

Old view machine inertia and nodal damping carry the margin	IBR view PLL, exciters, governors, stabilizers, inverter loops
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Classical weak-node or weak-link screens based on graph participation can identify a stressed place, but they may not identify the reason. The critical mode can belong to the control family, not the network family.

2. Why the scalar bridge fails

The standard reduced certificate puts damping into a fixed nodal coefficient D . In the studied ANDES cases,

$$\text{GENROU.D} = 0, \tag{1}$$

yet the full eigensolve has modal damping. The coefficient model lost the damping channel. The failure is not numerical. It is a modeling failure: controller states were condensed as if they had no frequency memory.

3. The controller-aware bridge

Partition the linearized ANDES state matrix into retained network/interface states and condensed control states. The reduced object is the Schur nonlinear eigenproblem

$$S_e(s) = sI - A_{ee} - A_{ec}(sI - A_{cc})^{-1}A_{ce}. \tag{2}$$

Equivalently, controller states enter the second-order view as a rational self-energy:

$$\Sigma(s) = D_0 + C(sI - A_{cc})^{-1}B. \tag{3}$$

The s -dependence is the point. It keeps controller memory in the reduced spectral problem.

4. What is classical, what is ours

Schur complements, Beyn contour extraction, nonlinear eigenvalue problems, and perturbation tools are classical. The contribution here is not new mathematics. The contribution is the applied formulation and audit: static condensation fails on the relevant cases, rational condensation recovers the critical poles, and the recovered pole family tells the engineer which lever to inspect.

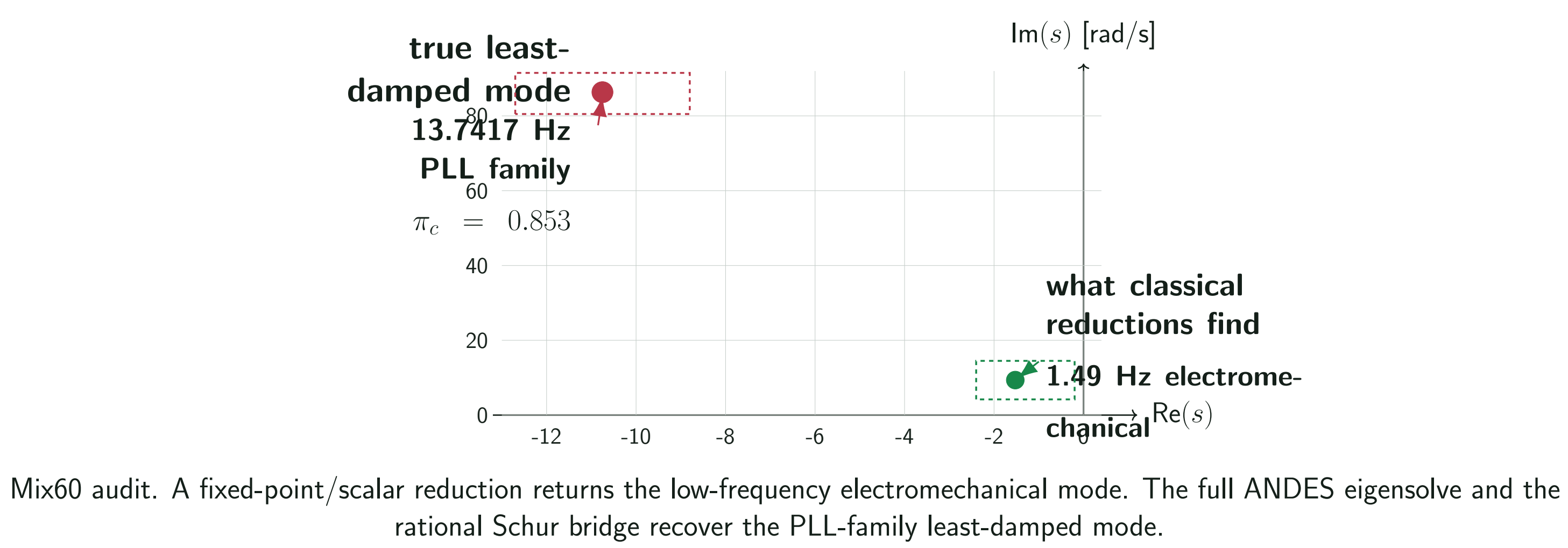
8. Scope guard

The bridge is validated to machine-level numerical agreement on diagnostic ANDES cases with generic controls. The controller-aware weak-element ranking is a formulation. Full ANDES validation over calibrated operating ranges remains in progress. The resonant weak-link mechanism is modal-local and control-conditional. It is not claimed as a dominant system-wide collapse mechanism.

9. How the poster sells to IAS

Problem: damping moved from machines into controls.
Risk: the standard reduced model can certify the wrong mode.
Result: the rational Schur bridge recovers the critical ANDES poles.
Action: decide whether the weakness is network, control, or resonant.

5. Dominant figure: you were looking in the wrong place



6. Validated bridge result

no-PSS critical pole matched	base critical pole matched	Mix60 PLL critical pole captured
$8.47 \times 10^{-11}\%$ worst frequency error		$2.48 \times 10^{-9}\%$ worst damping error

Beyn contour extraction of the Schur NEP matches full ANDES critical poles in the three diagnostic cases. This is machine-level numerical agreement for the bridge, not a claim that planning rankings are fully validated.

7. Utility: weak nodes and weak links

Network weak node reinforce topology; copper can move the mode	Control weak node retune PLL, inverter, governor, or stabilizer	Resonant weak link reinforcement can reduce damping
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The bridge classifies the weak element by family. That is the decision value: it tells the planner which lever is physically connected to the observed damping margin. In the line audit, 40 strict control-resonant records appeared among 348 global-margin-reversal records. Minority, but real.

Takeaway

Do not ask only where the grid is weak. Ask whether the weakness belongs to the network family or the control family. The fix is different.



Reproducibility
ANDES gates, NEP contour solver, and line-audit artifacts are included.
github.com/jlmayorgaco/paper-P06-spectral-graph-power-dynamics